

RC-IGBT TCAD Workbench Project

Public Release Documentation

This PDF documents the public GitHub release of the RC-IGBT TCAD Workbench project. The release is intentionally minimal: the visible project folder keeps only one sanitized ZIP package and one PDF document. Generated command decks, scripts, PLT/status files, and Workbench metadata are kept inside the ZIP package to avoid cluttering the repository browser.

Clean public package

training/RC-IGBT/	public release directory
RC-IGBT_swb_project_sanitized.zip	sanitized Workbench package
rc_igbt_project_documentation.pdf	PDF documentation

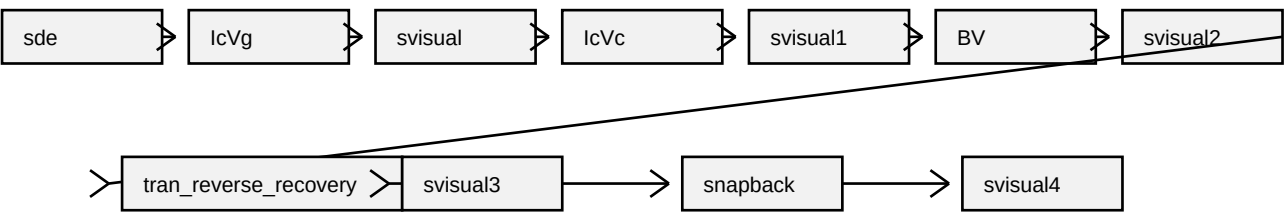
All scattered command/result/status files are kept inside the ZIP package only. Environment-specific parameters and binary states are not exposed in the visible folder.

Files kept in the visible project directory:

- RC-IGBT_swb_project_sanitized.zip
- rc_igbt_project_documentation.pdf

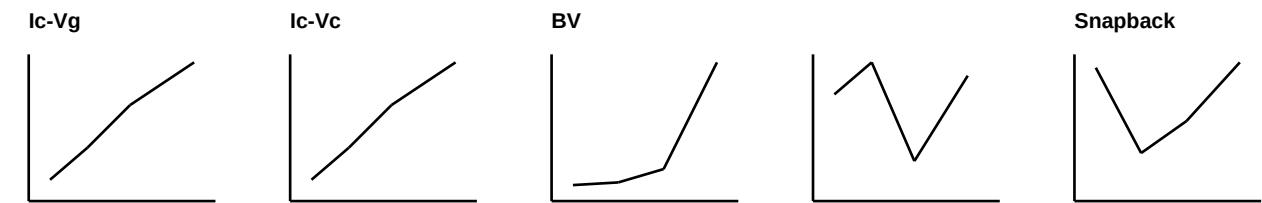
Everything else is stored inside the ZIP package or excluded for safety.

Workbench flow and simulation coverage



Main flow: SDE structure/mesh generation, Ic-Vg transfer curve, Ic-Vc output curve, breakdown simulation, reverse-recovery or double-pulse-style transient analysis, and snapback inspection. Node files, generated decks, and public-safe result files are packed inside the ZIP.

Result coverage



Simulation	Purpose	Expected output
Ic-Vg	gate control behavior	collector current vs gate voltage
Ic-Vc	output characteristic	collector current vs collector voltage
BV	blocking capability	
DPT / transient	switching and recovery	time-domain current/voltage

Snapback

snapback behavior

continuation/snapback curve

Public package policy and reproduction notes

Included in public package

OK - source .cmd decks
OK - Tcl/Python scripts
OK - PLT curve data
OK - status summaries
OK - selected figures
OK - sanitized ZIP

Not included in visible folder

NO - *.par material files
NO - environment files
NO - vendor models
NO - official example copies
NO - local VM/server paths
NO - TDR/SAV private paths

The public repository provides only a cleaned project package and a short PDF guide. Environment-specific resources and sensitive parameter data are not placed in the visible folder. Users should review the ZIP contents before reuse and provide local tool resources as needed.

Reproduction procedure

- Download RC-IGBT_swb_project_sanitized.zip from the GitHub project folder.
- Import or unpack it into a legal TCAD Workbench environment.
- Add local material parameter files if the decks reference external parameter files.
- Run structure/mesh generation before DC, transient, or snapback nodes.
- Compare generated curves with packaged PLT/status files for consistency.

Group	Examples	Review note
Geometry	cell pitch, drift thickness	check unit consistency
Doping	N-drift, P-body, emitter	do not publish confidential recipes
Physics	mobility, avalanche, thermal	keep model choices explicit
Electrodes	gate, emitter, collector	verify contact names
Solver settings	quasi/transient/continuation	important for convergence